Orbits

Notes on calculations on classical orbits under gravity

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Chapter 1

Central forces

1.1 Introduction

A central force is a force that acts only in the radial direction. In other words the force only depends on the distance from the centre. As such a central force can be defined like so

$$\mathbf{F} = f(r)\hat{\mathbf{r}} \tag{1.1.1}$$

The equation of motion for a central force is then

$$\ddot{\mathbf{r}} = \dot{\mathbf{v}} = \frac{\mathbf{F}}{m} = \frac{f(r)}{m}\hat{\mathbf{r}} \tag{1.1.2}$$

where we have used $\mathbf{v} = \dot{\mathbf{r}}$.

1.2 Angular Momentum

One important parameter for a central force is the angular momentum. The angular momentum is defined as the cross product of the position vector $\mathbf{r}$ and the momentum vector $\mathbf{p} = m\mathbf{v}$

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} = m\mathbf{r} \times \mathbf{v} \tag{1.2.1}$$

The mass m is fixed so it is more convenient to use the specific angular momentum

$$\mathbf{h} = \frac{\mathbf{L}}{m} = \mathbf{r} \times \mathbf{v} \tag{1.2.2}$$

1.2.1 Conservation of Angular Momentum

We can show that angular momentum is a constant of the motion by showing it's rate of change with respect to time is zero, ie it is fixed and unchanging. We start then by taking the derivative with respect to time.

$$\begin{aligned}
\dot{\mathbf{h}} &= \frac{d}{dt}(\mathbf{r} \times \mathbf{v}) \\
&= \dot{\mathbf{r}} \times \mathbf{v} + \mathbf{r} \times \dot{\mathbf{v}} \\
&= \mathbf{v} \times \mathbf{v} + \mathbf{r} \times \dot{\mathbf{v}} \\
&= \mathbf{r} \times \dot{\mathbf{v}}
\end{aligned} \tag{1.2.3}$$

We have used the fact that the vector product of any vector with itself is zero. We can now substitute in equation 1.1.2[1] for $\dot{\mathbf{v}}$

$$\begin{aligned}
\dot{\mathbf{h}} &= \mathbf{r} \times \dot{\mathbf{v}} \\
&= \mathbf{r} \times \frac{f(r)}{m}\hat{\mathbf{r}} \\
&= \frac{f(r)}{mr}\mathbf{r} \times \mathbf{r} \\
&= \mathbf{0}
\end{aligned} \tag{1.2.4}$$

This shows that the angular momentum is a constant of the motion for a central force.

[1]page 1

1.3 Potential Energy

The potential energy is the work done in moving the particle from a radius of ∞ to r. So the work done is given by

$$\begin{aligned} V(r) &= \int_\infty^r \mathbf{F}(r') \cdot d\mathbf{r}' \\ &= \int_\infty^r f(r')\hat{\mathbf{r}} \cdot dr'\hat{\mathbf{r}} \\ &= \int_\infty^r f(r')dr' \end{aligned} \tag{1.3.1}$$

if we let $U(x) = \int f(x)dx$ Then we get

$$V(r) = U(r) - U(\infty) \tag{1.3.2}$$

If we take the derivative of both sides with respect to r we get

$$\frac{dV}{dr} = \frac{d}{dr}(U(\infty) - U(r)) \tag{1.3.3}$$

1.4 Kinetic Energy

The kinetic energy of a particle is given by

$$T(v) = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v} = \frac{1}{2}mv^2 \tag{1.4.1}$$

1.5 Total Energy

The total energy is the sum of kinetic energy and potential energy

$$E = \frac{1}{2}mv^2 + \int_\infty^r f(r')dr' \tag{1.5.1}$$

Again it is often easier to use the specific total energy ϵ

$$\epsilon = \frac{E}{m} = \frac{1}{2}v^2 + \int_\infty^r \frac{f(r')}{m}dr' \tag{1.5.2}$$

to make this a little easier still we will let $\tilde{V} = \int_{\infty}^{r} \frac{f(r')}{m} dr'$, so we get

$$\epsilon = \frac{E}{m} = \frac{1}{2}v^2 + \tilde{V} \tag{1.5.3}$$

1.5.1 Conservation of Total Energy

As before with the angular momentum, we can show that the total energy is a constant of motion by taking the derivative with respect to time

$$\begin{aligned}
\frac{d\epsilon}{dt} &= \frac{d}{dt}\left(\frac{1}{2}v^2 + \tilde{V}\right) \\
&= \frac{1}{2}\frac{1}{2}\frac{dv^2}{dv}\frac{dv}{dt} + \frac{d\tilde{V}}{dr}\frac{dr}{dt} \\
&= \frac{1}{2}\frac{1}{2}(2v)\dot{v} + \frac{f(r)}{m}v \\
&= \frac{1}{2}v(\dot{v} + \frac{f(r)}{m})
\end{aligned} \tag{1.5.4}$$

Chapter 2

Gravity

Chapter 3

Geometry of an Ellipse

3.1 The String Construction of an Ellipse

One method of drawing an ellipse is to take a string of fixed length pinned to the two focus points of the ellipse. We can use this method to derive the standard equations for an ellipse. We will first start by setting up the geometry.

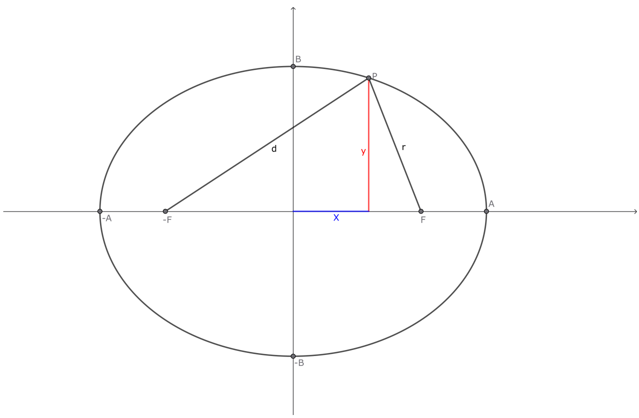

Figure 3.1: The string method of constructing an ellipse

We will work in a two dimensional Cartesian coordinate system with an ellipse centered at the origin and orientated so that it's semi-major axis is directed along the x-axis and it's semi-minor axis orientated with the y axis. The two focus points F and $-F$ are located on the x-axis a distance f away from the origin, so that $F = (f, 0)$ and $-F = (-f, 0)$. Let P be an arbitrary point on the ellipse with coordinates $P = (X, y)$[1]. The ellipse has semi-major axis length A and semi-minor axis length B, so that $-A \leq X \leq A$ and $-B \leq y \leq B$. Let r be the distance from F to P and d be the distance from $-F$ to P and let the sum of these be S, so that

$$S = r + d \tag{3.1.1}$$

where

$$\begin{aligned} r &= \sqrt{(X-f)^2 + y^2} \\ d &= \sqrt{(X+f)^2 + y^2} \end{aligned} \tag{3.1.2}$$

3.1.1 Special cases and useful results

From the construction method, we know that the sum is fixed. We can find it's exact value by considering P when $y = 0$ and $X = A$, so that

$$\begin{aligned} r &= \sqrt{(A-f)^2 + 0^2} \\ &= \sqrt{(A-f)^2} \\ &= A - f \\ d &= \sqrt{(A+f)^2 + 0^2} \\ &= \sqrt{(A+f)^2} \\ &= A + f \\ S &= r + d \\ &= A - f + A + f \\ S &= 2A \end{aligned} \tag{3.1.3}$$

[1]We use X rather than x here to avoid confusion with a later definition of x as a distance measured from the focus F

This shows that the string must be twice the length of the semi-major axis to work.

Another interesting result is obtained when we look at the point $P = (0, B)$

$$\begin{aligned}
r &= \sqrt{(0-f)^2 + B^2} \\
&= \sqrt{f^2 + B^2} \\
d &= \sqrt{(0+f)^2 + B^2} \\
&= \sqrt{(f^2 + B^2)^2} \\
S &= 2\sqrt{(f^2 + B^2)^2} \qquad (3.1.4)
\end{aligned}$$

since we now know that $S = 2A$ we can subtitute for S

$$\begin{aligned}
S &= 2\sqrt{(f^2 + B^2)^2} \\
2A &= 2\sqrt{(f^2 + B^2)^2} \\
A &= \sqrt{(f^2 + B^2)^2} \\
A^2 &= f^2 + B^2 \qquad (3.1.5)
\end{aligned}$$

You are more likely to see this stated as $B = A\sqrt{1-e^2}$ where e or the eccentricity is defined to be

$$e = \frac{f}{A} \qquad (3.1.6)$$

so that $f = eA$

3.1.2 Equation of an ellipse

Now we know the value of S we can look for a general equation that links X and y. Firstly we write out the sum in full

$$\begin{aligned}
S &= r + d \\
2A &= \sqrt{(X-f)^2 + y^2} + \sqrt{(X+f)^2 + y^2} \\
2A - \sqrt{(X-f)^2 + y^2} &= \sqrt{(X+f)^2 + y^2} \qquad (3.1.7)
\end{aligned}$$

Now we square both sides to get rid of the first square root

$$
\begin{aligned}
\left(2A - \sqrt{(X-f)^2 + y^2}\right)^2 &= (X+f)^2 + y^2 \\
4A^2 - 4A\sqrt{(X-f)^2 + y^2} + (X-f)^2 + \cancel{y^2} &= (X+f)^2 + \cancel{y^2} \\
4A^2 - 4A\sqrt{(X-f)^2 + y^2} + \cancel{X^2} - 2Xf + \cancel{f^2} &= \cancel{X^2} + 2Xf + \cancel{f^2} \\
4A^2 - 4A\sqrt{(X-f)^2 + y^2} &= 4Xf \\
A^2 - A\sqrt{(X-f)^2 + y^2} &= fX \\
A^2 - fX &= A\sqrt{(X-f)^2 + y^2} \\
A - \frac{fX}{A} &= \sqrt{(X-f)^2 + y^2}
\end{aligned}
\tag{3.1.8}
$$

Once again we square both sides to remove the square root

$$
\begin{aligned}
\left(A - \frac{fX}{A}\right)^2 &= (X-f)^2 + y^2 \\
A^2 - 2\cancel{A}\frac{fX}{\cancel{A}} + \left(\frac{f}{A}\right)^2 X^2 &= X^2 - 2fX + f^2 + y^2 \\
A^2 \cancel{-2fX} + \left(\frac{f}{A}\right)^2 X^2 &= X^2 \cancel{-2fX} + f^2 + y^2 \\
A^2 - f^2 &= \left(1 - \left(\frac{f}{A}\right)^2\right) X^2 + y^2 \\
&= \frac{A^2 - f^2}{A^2} X^2 + y^2 \\
1 &= \frac{X^2}{A^2} + \frac{y^2}{A^2 - f^2}
\end{aligned}
$$

This further simplifies if we use equation 3.1.5[2] which with some rearrangement gives $B^2 = A^2 - f^2$.

[2] page 9

And so we get the typical form of the equation of an ellipse

$$1 = \frac{X^2}{A^2} + \frac{y^2}{B^2} \tag{3.1.9}$$

3.2 Equation for an ellipse in polar form

Here we are going to look at the equation of an ellipse in polar coordinates with the positive focus at the origin. The closest point[3] lies along the positive x axis. This is the same system as we have been working with but shifted along the x axis so that the focus it at the origin, alternatively we can continue to use our centred ellipse but measure everything from the positive focus. This is in fact what we will do here and is also why we have made the distiction between the X coordinate and the x coordinate. Before looking at the relationship between X and x, we will look at the polar angle for an ellipse[4].

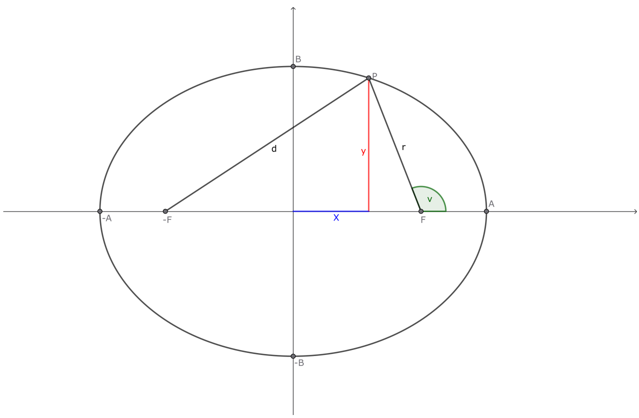

Figure 3.2: The polar angle for an ellipse

As you can see from Figure 3.2 we can determine the point P based on the polar angle ν

[3]Which as we will see later is called the periapsis for an orbit
[4]for orbits, this angle is called the true anomaly

$$
\begin{aligned}
X &= f + r\cos\nu \\
y &= r\sin\nu
\end{aligned}
\tag{3.2.1}
$$

we can also define x now also

$$
x = r\cos\nu \tag{3.2.2}
$$

Going back to the sum equation[5]

$$
\begin{aligned}
S &= r + d \\
2A - r &= d \\
(2A - r)^2 &= d^2 \\
(2A - r)^2 &= (X + f)^2 + y^2 \\
4A^2 - 4Ar + r^2 &= X^2 + 2Xf + f^2 + y^2 \\
4A^2 - 4Ar + r^2 &= X^2 + 2Xf + f^2 + y^2 \\
&= (f + r\cos\nu)^2 + 2f^2 + 2fr\cos\nu + f^2 + y^2 \\
&= f^2 + 2fr\cos\nu + r^2\cos^2\nu + 2f^2 + 2fr\cos\nu + f^2 + y^2 \\
&= 4f^2 + 4fr\cos\nu + r^2\cos^2\nu + y^2 \\
&= 4f^2 + 4fr\cos\nu + r^2\cos^2\nu + r^2\sin^2\nu \\
&= 4f^2 + 4fr\cos\nu + r^2 \\
4A^2 - 4Ar + \cancel{r^2} &= 4f^2 + 4fr\cos\nu + \cancel{r^2} \\
4A^2 - 4Ar &= 4f^2 + 4fr\cos\nu \\
A^2 - Ar &= f^2 + fr\cos\nu \\
A^2 - f^2 &= Ar + fr\cos\nu \\
A^2 - f^2 &= r(A + f\cos\nu) \\
r &= \frac{A^2 - f^2}{A + f\cos\nu}
\end{aligned}
\tag{3.2.3}
$$

We can do a little further simplifying by using Equation 3.1.6[6]

[5] Equation 3.1.3 on page 8

[6] page 9

$$
\begin{aligned}
r &= \frac{A^2 - f^2}{A + f\cos\nu} \\
&= \frac{A^2 - e^2A^2}{A + eA\cos\nu} \\
&= \frac{A^2(1-e^2)}{A(1+e\cos\nu)} \\
&= \frac{A(1-e^2)}{1+e\cos\nu} \qquad (3.2.4)
\end{aligned}
$$

Which is the standard form.

3.3 The Auxiliary Circle

Another parameterisation for an ellipse comes from looking at the relationship to a circle. In other words, an ellipse can also be seen as a flattened circle.

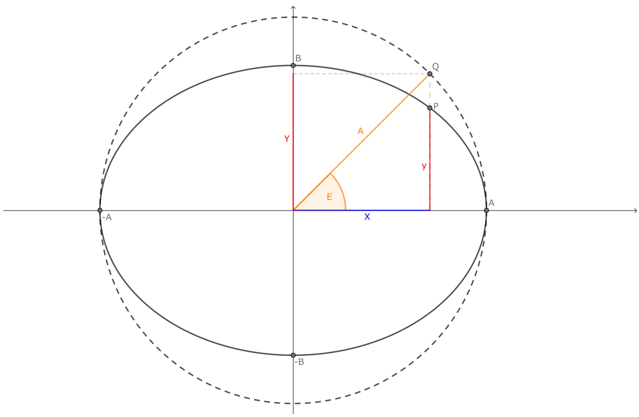

Figure 3.3: The ellipse and the auxiliary circle

Consider a circle with radius A which is centred on the ellipse with a semi-major axis of length A and a semi-minor axis of length B, as in Figure 3.3. The circle is known as the Auxiliary Circle.

We can define a point Q on the circle as $(A\cos E, A\sin E)$ so that the line from the origin to Q has an angle E[7] with the x axis.

We can now choose point P that is on the ellipse so that Q and P share the same x coordinate. We can work out the corresponding y coordinate by using the equation of an ellipse[8].

$$
\begin{aligned}
\frac{X^2}{A^2} + \frac{y^2}{B^2} &= 1 \\
\frac{A^2\cos^2 E}{A^2} + \frac{y^2}{B^2} &= 1 \\
\cos^2 E + \frac{y^2}{B^2} &= 1 \\
\frac{y^2}{B^2} &= 1 - \cos^2 E \\
\frac{y^2}{B^2} &= \sin^2 E \\
y^2 &= B^2 \sin^2 E \\
y &= B\sin E \qquad (3.3.1)
\end{aligned}
$$

Which leads to one of the standard parameterisations $P = (A\cos E, B\sin E)$

3.4 The Apses

The apses are the closest and furthest point from the positive focus of the ellipse. The periapsis is the closest point and the apoapsis is the farthest point. We will lable these p and a respectively.

As we can see from Figure 3.4 that the periapsis has a length of $p = A - f$ and the apoapsis has a length of $a = A + f$. We can put these in terms of the semi-major axis and the eccentricity by using equation 3.1.6[9]

[7]This angle is called the eccentric anomaly in terms of orbits

[8]Equation 3.1.9 on page 11

[9]page 9

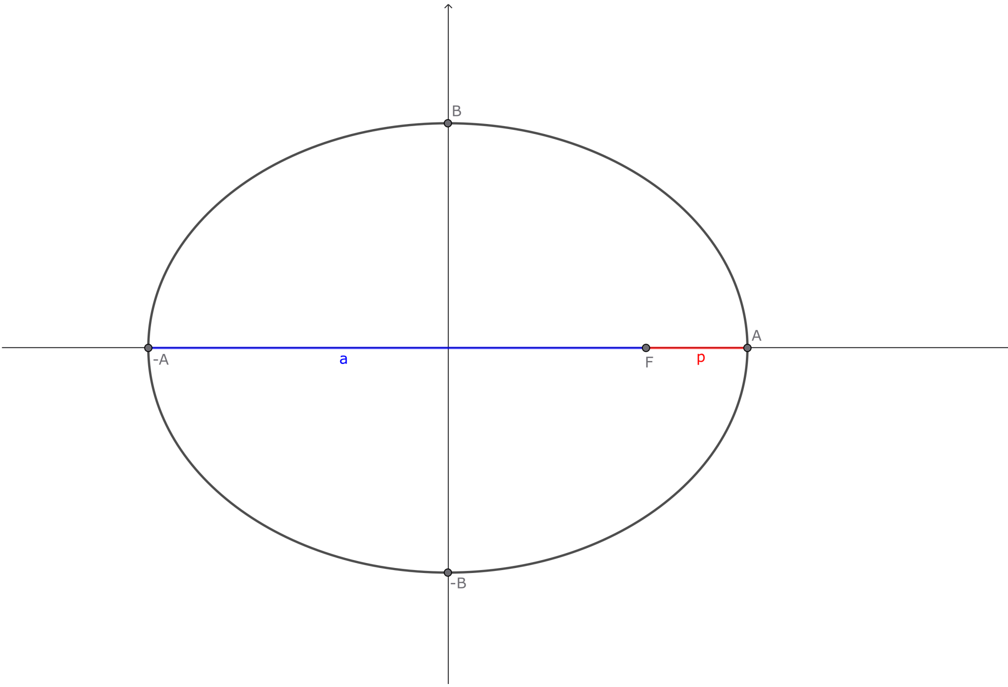

Figure 3.4: The ellipse with the apses highlighted

$$\begin{aligned} p &= A(1-e) \\ a &= A(1+e) \end{aligned} \tag{3.4.1}$$

We can also see that $A = p + a$ and that $f = a - p$

3.5 Relationship between true and eccentric anomalies

Here we will show that there is a relationship between the angles ν in Figure 3.2[10] and E in Figure 3.3[11]. In terms of orbits as we

[10] page 11
[11] page 13

will see later, ν is called the true anomaly and E is the eccentric anomaly.

We start by looking at the tangent of ν and using equation 3.1.5[12] in particular the form $B = A\sqrt{1-e^2}$ and equation 3.1.6[13] in particular the form $f = eA$. Finally we need equations 3.2.1[14] and 3.2.2[15].

$$\begin{aligned} x &= r\cos\nu \\ &= A\cos E - f \\ &= A\cos E - eA \\ &= A(\cos E - e) \end{aligned} \tag{3.5.1}$$

$$\begin{aligned} y &= r\sin\nu \\ &= B\sin E \\ &= A\sqrt{1-e^2}\sin E \end{aligned} \tag{3.5.2}$$

We can now put these into

$$\begin{aligned} \tan\nu &= \frac{y}{x} \\ &= \frac{A\sqrt{1-e^2}\sin E}{A(\cos E - e)} \\ &= \frac{\sqrt{1-e^2}\sin E}{\cos E - e} \end{aligned} \tag{3.5.3}$$

Here we can use the half tan formulas A.1.1[16] and A.1.2[17] to

[12] page 9
[13] page 9
[14] page 12
[15] page 12
[16] page 27
[17] page 27

replace the sine and cosine

$$\begin{aligned}
\tan\nu &= \frac{\sqrt{1-e^2}\sin E}{\cos E - e} \\
&= \frac{\sqrt{1-e^2}\frac{2\tan\frac{E}{2}}{1+\tan^2\frac{E}{2}}}{\frac{1-\tan^2\frac{E}{2}}{1+\tan^2\frac{E}{2}} - e}
\end{aligned} \tag{3.5.4}$$

Because the next steps can get a little messy, we'll introduce some variables to help, let $t = \tan\frac{E}{2}$

$$\begin{aligned}
\tan\nu &= \frac{\sqrt{1-e^2}\frac{2\tan\frac{E}{2}}{1+\tan^2\frac{E}{2}}}{\frac{1-\tan^2\frac{E}{2}}{1+\tan^2\frac{E}{2}} - e} \\
&= \frac{\sqrt{1-e^2}\frac{2t}{1+t^2}}{\frac{1-t^2}{1+t^2} - e} \\
&= \frac{\sqrt{1-e^2}\frac{2t}{1+t^2}}{\frac{1-t^2-e-et^2}{1+t^2}} \\
&= \frac{\sqrt{1-e^2}2t}{1-t^2-e-et^2} \\
&= \frac{\sqrt{1-e^2}2t}{1-e-(1+e)t^2} \\
&= \frac{\sqrt{1-e^2}2t}{(1+e)\left(\frac{1-e}{1+e}-t^2\right)} \\
&= \frac{\sqrt{(1-e)(1+e)}2t}{(1+e)\left(\frac{1-e}{1+e}-t^2\right)} \\
&= \sqrt{\frac{1-e}{1+e}}\frac{2t}{\left(\sqrt{\frac{1-e}{1+e}}\right)^2 - t^2}
\end{aligned} \tag{3.5.5}$$

Again for ease, let $\alpha = \sqrt{\frac{1-e}{1+e}}$

$$\begin{aligned}
\tan\nu &= \sqrt{\frac{1-e}{1+e}}\frac{2t}{\left(\sqrt{\frac{1-e}{1+e}}\right)^2 - t^2} \\
&= \alpha\frac{2t}{\alpha^2 - t^2} \\
&= \alpha\frac{2t}{\alpha^2\left(1 - \frac{t^2}{\alpha^2}\right)} \\
&= \frac{2\frac{t}{\alpha}}{\left(1 - \frac{t^2}{\alpha^2}\right)}
\end{aligned} \tag{3.5.6}$$

Now we now use Equation A.1.3[18] to replace $\tan\nu$ with $\tan\frac{\nu}{2}$

$$\begin{aligned}
\tan\nu &= \frac{2\frac{t}{\alpha}}{\left(1 - \frac{t^2}{\alpha^2}\right)} \\
\frac{2\tan\frac{\nu}{2}}{1-\tan^2\frac{\nu}{2}} &= \frac{2\frac{t}{\alpha}}{\left(1 - \frac{t^2}{\alpha^2}\right)}
\end{aligned} \tag{3.5.7}$$

you can see that

$$\tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2} \tag{3.5.8}$$

[18] page 27

Chapter 4

Perifocal Plane

Chapter 5

Elliptical Orbits

Chapter 6

Kepler's Equation

Chapter 7

Initial Conditions

7.1 Specific Total Energy

7.2 Specific Angular Momentum

7.3 Eccentricity Vector

7.4 Direction of Perifocal Plane

7.5 Phase of the Orbit, M_0

7.6 Time Evolution of the Position

Appendix A

Useful Identities

A.1 Trigonometric

$$\sin\alpha = \frac{2\tan\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}} \tag{A.1.1}$$

$$\cos\alpha = \frac{1-\tan^2\frac{\alpha}{2}}{1+\tan^2\frac{\alpha}{2}} \tag{A.1.2}$$

$$\tan\alpha = \frac{2\tan\frac{\alpha}{2}}{1-\tan^2\frac{\alpha}{2}} \tag{A.1.3}$$

A.2 Vectors

Appendix B

Old Stuff to review

B.1 Plane Polar Coordinate System

In this section we will derive the equations of motion for an object acted upon by gravity in plane polar coordinates. To do this we will first consider plane polar coordinates themselves and the derivatives of the position vector within the coordinate system.

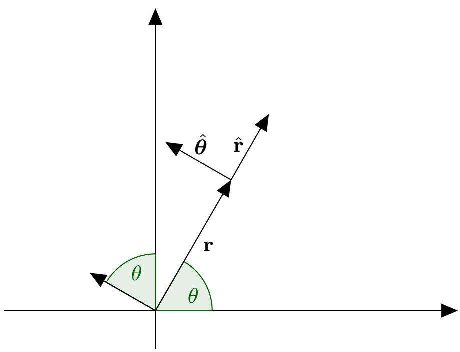

Figure B.1: Position and unit vectors in plane polar coordinates

B.1.1 Polar unit vectors

The unit vectors of plane polar coordinates can be determined from the standard unit vectors as

$$\hat{\boldsymbol{r}} = \hat{\mathbf{\imath}}\cos(\theta) + \hat{\mathbf{\jmath}}\sin(\theta) \tag{B.1.1}$$

$$\hat{\boldsymbol{\theta}} = -\hat{\mathbf{\imath}}\sin(\theta) + \hat{\mathbf{\jmath}}\cos(\theta) \tag{B.1.2}$$

Where r is the distance from the origin to the position or alternatively the length of the position vector, and θ is the angle between $\hat{\mathbf{\imath}}$ and $\boldsymbol{r}$. Since the unit vectors $\hat{\boldsymbol{r}}$ and $\hat{\boldsymbol{\theta}}$ depend on θ which changes with time, the unit vectors also change with time. We will begin by looking at the derivative of $\hat{\boldsymbol{r}}$ with respect to time.

$$\frac{d\hat{\boldsymbol{r}}}{dt} = \frac{d\theta}{dt}\frac{d}{d\theta}\hat{\boldsymbol{r}} = \frac{d\theta}{dt}\left[-\hat{\mathbf{\imath}}\sin(\theta) + \hat{\mathbf{\jmath}}\cos(\theta)\right] = \frac{d\theta}{dt}\hat{\boldsymbol{\theta}} \tag{B.1.3}$$

where we have used equation B.1.2 to replace the typical Cartesian coordinate unit vectors.

$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d\theta}{dt}\frac{d}{dt}\hat{\boldsymbol{\theta}} = \frac{d\theta}{dt}\left[-\hat{\mathbf{\imath}}\cos(\theta) - \hat{\mathbf{\jmath}}\sin(\theta)\right] = -\frac{d\theta}{dt}\hat{\boldsymbol{r}} \tag{B.1.4}$$

Since $\frac{d\hat{\boldsymbol{r}}}{dt}$ too depends on θ so is also time dependent.

$$\frac{d^2\hat{\boldsymbol{r}}}{dt^2} = \frac{d}{dt}\left[\frac{d\theta}{dt}\hat{\boldsymbol{\theta}}\right] = \frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} + \frac{d\theta}{dt}\frac{d\hat{\boldsymbol{\theta}}}{dt} = \frac{d^2\theta}{dt^2}\hat{\boldsymbol{\theta}} - \left(\frac{d\theta}{dt}\right)^2\hat{\boldsymbol{r}} \tag{B.1.5}$$

Polar unit vectors summary

The polar coordinate unit vectors are

$$\hat{\boldsymbol{r}} = \hat{\mathbf{\imath}}\cos(\theta) + \hat{\mathbf{\jmath}}\sin(\theta)$$

$$\hat{\boldsymbol{\theta}} = -\hat{\mathbf{\imath}}\sin(\theta) + \hat{\mathbf{\jmath}}\cos(\theta)$$

With the first derivatives being

$$\frac{d\hat{\boldsymbol{r}}}{dt} = \omega\hat{\boldsymbol{\theta}}$$
$$\frac{d\hat{\boldsymbol{\theta}}}{dt} = -\omega\hat{\boldsymbol{r}}$$

And the second derivative of $\hat{\boldsymbol{r}}$ being

$$\frac{d^2\hat{\boldsymbol{r}}}{dt^2} = \frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - \omega^2\hat{\boldsymbol{r}}$$

with $\omega = \frac{d\theta}{dt}$

B.1.2 Velocity and acceleration in polar coordinates

Here we will calculate the acceleration in terms of the polar unit vectors.

Position

Starting with the position vector

$$\mathbf{r} = r\hat{\mathbf{r}} \tag{B.1.6}$$

This is the length of the position vector in the direction of $\hat{\boldsymbol{r}}$.

Velocity

The first derivative will be the velocity of the position vector in terms of the polar unit vectors

$$\frac{d\boldsymbol{r}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\frac{d\hat{\boldsymbol{r}}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}} \tag{B.1.7}$$

The velocity is the sum of the radial velocity in the radial direction $\dot{r}$ and the angular part of the velocity $r\omega$ in the angular direction.

Acceleration

$$\begin{aligned}\frac{d^2\boldsymbol{r}}{dt^2} &= \frac{d^2r}{dt^2}\hat{\boldsymbol{r}} + \frac{dr}{dt}\frac{d\hat{\boldsymbol{r}}}{dt} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} + r\omega\frac{d\hat{\boldsymbol{\theta}}}{dt} \\ &= \frac{d^2r}{dt^2}\hat{\boldsymbol{r}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + \frac{dr}{dt}\omega\hat{\boldsymbol{\theta}} + r\frac{d\omega}{dt}\hat{\boldsymbol{\theta}} - r\omega^2\hat{\boldsymbol{r}} \\ &= \left[\frac{d^2r}{dt^2} - r\omega^2\right]\hat{\boldsymbol{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right]\hat{\boldsymbol{\theta}} \end{aligned} \tag{B.1.8}$$

Acceleration in polar coordinates summary

The position vector

$$\boldsymbol{r} = r\hat{\boldsymbol{r}}$$

Velocity

$$\frac{d\boldsymbol{r}}{dt} = \frac{dr}{dt}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}$$

Acceleration

$$\frac{d^2\boldsymbol{r}}{dt^2} = \left[\frac{d^2r}{dt^2} - r\omega^2\right]\hat{\boldsymbol{r}} + \left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right]\hat{\boldsymbol{\theta}}$$

B.2 Gravity

B.2.1 Gravity as a central force

The force between two particles a distance r apart with mass M and m respectivly is given by

$$\boldsymbol{F} = -\frac{GMm}{r^2}\hat{\boldsymbol{r}} \tag{B.2.1}$$

The acceleration on the particle of mass m due to the force of gravity between the two particles is

$$\ddot{\boldsymbol{r}} = \frac{\boldsymbol{F}}{m} = -\frac{GM}{r^2}\hat{\boldsymbol{r}} \tag{B.2.2}$$

Using equation B.1.8 we can split the acceleration into radial and angular parts

$$-\frac{GM}{r^2} = \ddot{r} - r\omega^2 \tag{B.2.3}$$

$$0 = r\ddot{\theta} + 2\dot{r}\dot{\theta} \tag{B.2.4}$$

B.3 Angular Momentum

The angular momentum of a particle is given by

$$\begin{aligned}
\boldsymbol{L} &= \boldsymbol{r} \times \boldsymbol{p} \\
&= m\boldsymbol{r} \times \frac{d\boldsymbol{r}}{dt} \\
&= mr\hat{\boldsymbol{r}} \times \left[\dot{r}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}\right] \\
&= m\left[r\dot{r}\hat{\boldsymbol{r}} \times \hat{\boldsymbol{r}} + r^2\omega\hat{\boldsymbol{r}} \times \hat{\boldsymbol{\theta}}\right] \\
&= mr^2\omega\hat{\boldsymbol{k}} \qquad \text{(B.3.1)}
\end{aligned}$$

We can also write this as $\mathbf{L} = l\hat{\boldsymbol{k}}$ were

$$l = mr^2\omega \qquad \text{(B.3.2)}$$

B.3.1 Conservation Of Angular Momentum

We can show now that angular momentum is conserved under a central force by taking the derivative with respect to time.

$$\frac{d\mathbf{L}}{dt} = \frac{dl\hat{\boldsymbol{k}}}{dt} = \frac{dl}{dt}\hat{\boldsymbol{k}} \qquad \text{(B.3.3)}$$

$$\begin{aligned}
\frac{dl}{dt} &= m\frac{d}{dt}\left[r^2\omega\right] \\
&= m\left[\frac{dr^2}{dt}\omega + r^2\frac{d\omega}{dt}\right] \\
&= m\left[\frac{dr^2}{dr}\frac{dr}{dt}\omega + r^2\frac{d\omega}{dt}\right] \\
&= m\left[2r\frac{dr}{dt}\omega + r^2\frac{d\omega}{dt}\right] \\
&= mr\left[2\frac{dr}{dt}\omega + r\frac{d\omega}{dt}\right] \qquad \text{(B.3.4)}
\end{aligned}$$

The part inside the brackets is the angular acceleration which for a central force is zero[1] and so

$$\begin{aligned} \frac{dl}{dt} &= mr \times 0 \\ &= 0 \end{aligned} \tag{B.3.5}$$

Which means that the angular momentum is constant.

[1]see equation B.2.4

B.4 Energy

B.4.1 Potential Energy

The potential energy is the work done in moving the particle from a radius of ∞ to r. So the work done is given by

$$
\begin{aligned}
U(r) &= \int_{\infty}^{r} \boldsymbol{F}(r') \cdot d\boldsymbol{r}' \\
U(r) &= \int_{\infty}^{r} -\frac{GMm}{r'^2}\hat{\boldsymbol{r}} \cdot dr'\hat{\boldsymbol{r}} \\
U(r) &= -GMm \int_{\infty}^{r} \frac{dr'}{r'^2} \\
U(r) &= -GMm \left[\frac{-1}{r'}\right]_{\infty}^{r} \\
U(r) &= GMm \left[\frac{1}{\infty} - \frac{1}{r}\right] \\
U(r) &= -\frac{GMm}{r} \qquad \text{(B.4.1)}
\end{aligned}
$$

B.4.2 Kinetic Energy

The kinetic energy of a particle is given by

$$
T(\dot{r}) = \frac{1}{2}m\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} \qquad \text{(B.4.2)}
$$

using eqation B.1.2 gives

$$
\begin{aligned}
T(\dot{r}) &= \frac{1}{2}m\left[\dot{r}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}\right] \cdot \left[\dot{r}\hat{\boldsymbol{r}} + r\omega\hat{\boldsymbol{\theta}}\right] \\
&= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\omega^2 \qquad \text{(B.4.3)}
\end{aligned}
$$

we can also make the subsitution using using the magnitude of the angular momentum[2] $l = mr^2\omega$

[2] equation B.3.2

$$
\begin{aligned}
l &= mr^2\omega \\
\omega &= \frac{l}{mr^2}
\end{aligned}
\tag{B.4.4}
$$

using equation B.4.4 in equation B.4.3 leads to

$$
\begin{aligned}
T(\dot{r}) &= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\omega^2 \\
&= \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\frac{l^2}{m^2r^4} \\
&= \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2}
\end{aligned}
\tag{B.4.5}
$$

Total Energy

Finally the total energy can be found by putting together the equation for kinetic energy and potential energy

$$
\begin{aligned}
E(r,\dot{r}) &= T(\dot{r}) + U(r) \\
E(r,\dot{r}) &= \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}
\end{aligned}
\tag{B.4.6}
$$

sometimes the last two terms of equation B.4.6 are put together to form a new potential energy type term that is dependent only on the radius and the angular momentum.

B.4.3 Conservation of Total Energy

Just like with angular momentum we can show that total energy is conserved by taking the derivative with respect to time.

$$\begin{aligned}
\frac{dE}{dt} &= \frac{d}{dt}\left[\frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}\right] \\
&= \frac{1}{2}m\frac{d\dot{r}^2}{dt} + \frac{l^2}{2m}\frac{dr^{-2}}{dt} - GMm\frac{dr^{-1}}{dt} \\
&= \frac{1}{2}m\frac{d\dot{r}^2}{d\dot{r}}\frac{d\dot{r}}{dt} + \frac{l^2}{2m}\frac{dr^{-2}}{dr}\frac{dr}{dt} - GMm\frac{dr^{-1}}{dr}\frac{dr}{dt} \\
&= \frac{1}{2}m(2\dot{r})\ddot{r} + \frac{l^2}{2m}\left(\frac{-2}{r^3}\right)\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\ddot{r} - \frac{l^2}{mr^3}\dot{r} + \frac{GMm}{r^2}\dot{r} \qquad \text{(B.4.7)}
\end{aligned}$$

using the magnitude of the angular momentum[3] $l = mr^2\omega$

$$\begin{aligned}
\frac{dE}{dt} &= m\dot{r}\ddot{r} - \frac{l^2}{mr^3}\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\ddot{r} - \frac{m^2r^4\omega^2}{mr^3}\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\ddot{r} - mr\omega^2\dot{r} + \frac{GMm}{r^2}\dot{r} \\
&= m\dot{r}\left[\ddot{r} - r\omega^2 + \frac{GM}{r^2}\right] \qquad \text{(B.4.8)}
\end{aligned}$$

using the radial part of the acceleration[4] $-\frac{GM}{r^2} = \ddot{r} - r\omega^2$ we get

$$\begin{aligned}
\frac{dE}{dt} &= m\dot{r}\left[-\frac{GM}{r^2} + \frac{GM}{r^2}\right] \\
&= m\dot{r}\,[0] \\
&= 0 \qquad \text{(B.4.9)}
\end{aligned}$$

This then shows that the total energy does not change with respect to time, so is a constant and also conserved.

[3] equation B.3.2
[4] equation B.2.3

B.5 Geometry Of Elipses

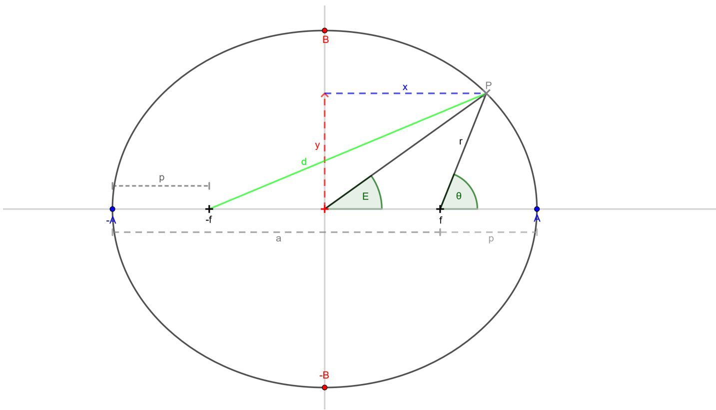

Figure B.2: An Elipse with important values shown

In figure B.2 a point P on the ellipse forms an angle E with the origin and the x axis, this angle is called the *Eccentric Anomaly*.

Also the angle θ is formed at the ellipse's focus between the point on the ellipse and the axis.

We also define the distances a and p as the apoapsis and periapsis distances respectively.

The semi major axis has been labeled as A, so that the total width of the ellipse is $2A$. Also shown on the diagram is the semi minor axis B, so the total height of the ellipse is $2B$.

For this text we will take the elliptical eccentricity to be defined as the ratio of the focus and the semi major axis.

$$e = \frac{f}{A} \tag{B.5.1}$$

B.5.1 The Apses

Before we get any further, we will look at the apses. From figure B.2 we can see that

$$p = A - f = A - eA = A(1 - e) \tag{B.5.2}$$

and that

$$a = A + f = A + eA = A(1 + e) \tag{B.5.3}$$

also

$$a + p = A - f + A + f = 2A \tag{B.5.4}$$
$$a - p = A + f - A + f = 2f \tag{B.5.5}$$

which can give us the eccentricity as defined by the apses

$$e = \frac{f}{A} = \frac{2f}{2A} = \frac{a - p}{a + p} \tag{B.5.6}$$

B.5.2 Sum of distances from foci and P

One property of an ellipse is that the sum of the distances from any point on the ellipse and the foci is a constant. This fact allows you to draw an ellipse using two pins and a piece of string. We will use this fact and look at two special cases, before looking at the general case and deriving the equation for an ellipse. Before we do we will define what the sum is here.

$$S = d + r \tag{B.5.7}$$

Special Case 1: Point on the x axis When the point is at $(A, 0)$. The distance d is $f + A$ and the distance r is $A - f$ from the diagram above. So our sum becomes

$$S = d + r = f + A + A - f = 2A \tag{B.5.8}$$

Special Case 2: Point on the y axis When the point is on the y axis, it is at the point $(0, B)$. The distance d is $\sqrt{f^2 + B^2}$ and the distance r is also $\sqrt{f^2 + B^2}$ from the diagram above. So our sum is

$$S = 2\sqrt{f^2 + B^2} \tag{B.5.9}$$

and if the sum is a constant then this will be equal to the value obtained before[5].

$$S = 2\sqrt{f^2 + B^2} = 2A \tag{B.5.10}$$

and so we now have the relationship between the semi major and minor axes and the focus

$$A^2 = f^2 + B^2 \tag{B.5.11}$$

This can also be used to get B in terms of A using the connection to the eccentricity[6]

$$\begin{aligned} B^2 &= A^2 - f^2 \\ &= A^2 - (eA)^2 \\ &= A^2(1 - e^2) \\ &\Rightarrow B = A\sqrt{1 - e^2} \end{aligned} \tag{B.5.12}$$

additionally, if we use the identity $x^2 - y^2 = (x + y)(x - y)$ we can also get the following

$$B = A\sqrt{1 - e^2} = \sqrt{A(1 + e) \times A(1 - e)} = \sqrt{ap} \tag{B.5.13}$$

where we have used equations B.5.2[7] and B.5.3[8] to substitute in for a and b respectively.

General case In general for a point at (x, y) we get the following

$$r^2 = (x - f)^2 + y^2 \tag{B.5.14}$$

$$d^2 = (x + f)^2 + y^2 \tag{B.5.15}$$

by looking at the triangles formed in figure B.2 with hypotenuse being r and d respectively. The sum[9] then becomes

$$\begin{aligned} S = 2A &= \sqrt{(x - f)^2 + y^2} + \sqrt{(x + f)^2 + y^2} \\ 2A - \sqrt{(x - f)^2 + y^2} &= \sqrt{(x + f)^2 + y^2} \end{aligned} \tag{B.5.16}$$

[5]equation B.5.8 on page **??**

[6]equation B.5.1 on page 39

[7]on page 39

[8]on page 40

[9]equation B.5.7 on page 40

squaring both sides gives

$$\begin{aligned}
\left(2A - \sqrt{(x-f)^2+y^2}\right)^2 &= (x+f)^2+y^2 \\
4A^2 - 4A\sqrt{(x-f)^2+y^2} + (x-f)^2 + y^2 &= (x+f)^2+y^2 \\
4A^2 - 4A\sqrt{(x-f)^2+y^2} + x^2 - 2xf + f^2 &= x^2 + 2xf + f^2 \\
4A^2 - 4A\sqrt{(x-f)^2+y^2} &= 4xf \\
4A^2 - 4xf &= 4A\sqrt{(x-f)^2+y^2} \\
\frac{4A^2 - 4xf}{4A} &= \sqrt{(x-f)^2+y^2} \\
\sqrt{(x-f)^2+y^2} &= A - \frac{xf}{A} \qquad \text{(B.5.17)}
\end{aligned}$$

Squaring both sides gives again gives

$$\begin{aligned}
(x-f)^2+y^2 &= \left(A - \frac{xf}{A}\right)^2 \\
x^2 - 2xf + f^2 + y^2 &= A^2 - 2\frac{xfA}{A} + \frac{x^2f^2}{A^2} \\
x^2 - 2xf + f^2 + y^2 &= A^2 - 2xf + e^2x^2 \\
(1-e^2)x^2 + y^2 &= A^2 - f^2 \\
(1-e^2)x^2 + y^2 &= A^2(1-e^2) \\
\frac{(1-e^2)x^2}{A^2(1-e^2)} + \frac{y^2}{A^2(1-e^2)} &= 1 \\
\frac{x^2}{A^2} + \frac{y^2}{B^2} &= 1 \qquad \text{(B.5.18)}
\end{aligned}$$

Which is the cartesian form of an elipse, there by showing that the elipse does indeed obey the contstraint that the sum of the distances from the foci is a constant.

Polar Form Once again returning to figure B.2 we can see that

$$x = f + r\cos\theta \qquad \text{(B.5.19)}$$

$$y = r\sin\theta \qquad \text{(B.5.20)}$$

The sum[10] then becomes

$$\begin{aligned} S = 2A &= r + d \\ 2A - r &= d \end{aligned} \tag{B.5.21}$$

We can square both sides to get

$$\begin{aligned} (2A - r)^2 &= d^2 \\ 4A^2 - 4Ar + r^2 &= (x + f)^2 + y^2 \\ &= x^2 + 2fx + f^2 + y^2 \\ &= (f + r\cos\theta)^2 + 2fx + f^2 + r^2\sin^2\theta \\ &= f^2 + 2fr\cos\theta + r^2\cos^2\theta + 2f\,(f + r\cos\theta) + f^2 \\ &+ r^2\sin^2\theta \\ &= f^2 + 2fr\cos\theta + 2f^2 + 2fr\cos\theta + f^2 \\ &+ r^2(\sin^2\theta + \cos^2\theta) \\ &= 4f^2 + 4fr\cos\theta + r^2 \\ A^2 - Ar &= f^2 + fr\cos\theta \\ A^2 - f^2 &= (A + f\cos\theta)\,r \\ A^2 - e^2A^2 &= (A + eA\cos\theta)\,r \\ A^2\left(1 - e^2\right) &= A\,(1 + e\cos\theta)\,r \\ A\left(1 - e^2\right) &= (1 + e\cos\theta)\,r \end{aligned} \tag{B.5.22}$$

Which with some rearranging results in the polar form for the equation of an elipse

$$r = \frac{A\,(1 - e^2)}{1 + e\cos\theta} \tag{B.5.23}$$

[10]equation B.5.7 40

B.5.3 Relationship between the distance from the focus and the Eccentric Anomaly

Looking at figure B.2 again we can see that x and y can also be given by

$$x = A\cos E \tag{B.5.24}$$

$$y = B\sin E = A\sqrt{1-e^2}\sin E \tag{B.5.25}$$

We can also see that r can be given by

$$\begin{aligned}
r^2 &= (x-f)^2 + y^2 \\
&= x^2 - 2xf + f^2 + y^2 \\
&= A^2\cos^2 E - 2eA^2\cos E + e^2A^2 + A^2(1-e^2)\sin^2 E \\
\left(\frac{r}{A}\right)^2 &= \cos^2 E - 2e\cos E + e^2 + (1-e^2)\sin^2 E \\
\left(\frac{r}{A}\right)^2 &= \cos^2 E - 2e\cos E + e^2 + \sin^2 E - e^2\sin^2 E \\
\left(\frac{r}{A}\right)^2 &= \left(\sin^2 E + \cos^2 E\right) - 2e\cos E + e^2(1-\sin^2 E) \\
\left(\frac{r}{A}\right)^2 &= 1 - 2e\cos E + e^2\cos^2 E \\
\left(\frac{r}{A}\right)^2 &= (1)^2 - 2(1)(e\cos E) + (e\cos E)^2 \\
\left(\frac{r}{A}\right)^2 &= (1 - e\cos E)^2 \qquad \text{(B.5.26)}
\end{aligned}$$

leading to

$$r = A\,(1 - e\cos E) \tag{B.5.27}$$

Summary

$$p = A(1-e)$$

$$a = A(1+e)$$

$$f = eA$$

$$A = \frac{a+p}{2}$$

$$f = \frac{a-p}{2}$$

$$e = \frac{a-p}{a+p}$$

$$\begin{aligned} B^2 &= A^2 - f^2 \\ &= A^2(1-e^2) \\ &= A(1+e)A(1-e) \\ &= ap \\ B &= \sqrt{ap} \end{aligned}$$

$$\begin{aligned} r &= \frac{A(1-e^2)}{1+e\cos\theta} \\ &= A\left(1-e\cos E\right) \end{aligned}$$

$$\begin{aligned} x &= A\cos E \\ &= f + r\cos\theta \end{aligned}$$

$$\begin{aligned} y &= B\sin E \\ &= r\sin\theta \end{aligned}$$

B.6 Kepler's Equation

Starting with equation B.5.27 that gives the relationship between the distance from the focus and the ecentric anomaly

$$r = A(1 - e\cos E)$$

We can take the derivative with respect to time to get the relationship between the rate of change of distance with time and the ecentric anomally.

$$\begin{aligned}
\frac{dr}{dt} &= A\frac{d}{dt}\left(1 - e\cos E\right)\\
&= A\frac{dE}{dt}\frac{d}{dE}\left(1 - e\cos E\right)\\
&= eA\sin E\frac{dE}{dt}
\end{aligned} \tag{B.6.1}$$

Using equation B.5.23 for the equation of an elipse in polar form $r = \frac{A\left(1-e^2\right)}{1+e\cos\theta}$. We can invert this this to get

$$\frac{1}{r} = \frac{1 + e\cos\theta}{A\left(1 - e^2\right)} \tag{B.6.2}$$

which will make taking the derivative with respect to time easier.

$$\begin{aligned}
\frac{d}{dt}\left(\frac{1}{r}\right) &= \frac{d}{dt}\left[\frac{1 + e\cos\theta}{A\left(1 - e^2\right)}\right]\\
\frac{dr}{dt}\frac{d}{dr}\left(\frac{1}{r}\right) &= \frac{1}{A(1-e^2)}\frac{d\theta}{dt}\frac{d}{d\theta}\left[1 + e\cos\theta\right]\\
-\frac{1}{r^2}\frac{dr}{dt} &= \frac{-e\sin\theta}{A(1-e^2)}\frac{d\theta}{dt}
\end{aligned} \tag{B.6.3}$$

we can now substitute in equation B.6.1 for $\frac{dr}{dt}$ to get

$$-\frac{1}{r^2}eA\sin E\frac{dE}{dt} = \frac{-e\sin\theta}{A(1-e^2)}\frac{d\theta}{dt} \tag{B.6.4}$$

We can use equation B.5.20 and B.5.25 for y to obtain a relationship between $\sin\theta$ and $\sin E$

$$y = r\sin\theta = A\sqrt{1-e^2}\sin E \tag{B.6.5}$$

We can use this relationship to replace the $\sin\theta$ term in equation B.6.4

$$\begin{aligned}
-\frac{1}{r^2}eA\sin E\frac{dE}{dt} &= \frac{-e\sin\theta}{A(1-e^2)}\frac{d\theta}{dt} \\
&= \frac{-eA\sqrt{1-e^2}\sin E}{Ar(1-e^2)}\frac{d\theta}{dt} \\
&= \frac{-e\sin E}{r\sqrt{1-e^2}}\frac{d\theta}{dt} \\
\frac{-eA\sin E}{r^2}\frac{dE}{dt} &= \frac{-e\sin E}{r\sqrt{1-e^2}}\frac{d\theta}{dt} \\
\frac{dE}{dt} &= \frac{-r^2}{eA\sin E}\frac{-e\sin E}{r\sqrt{1-e^2}}\frac{d\theta}{dt} \\
\frac{dE}{dt} &= \frac{r}{A\sqrt{1-e^2}}\frac{d\theta}{dt} \qquad \text{(B.6.6)}
\end{aligned}$$

Finally we can replace r with equation B.5.27 $r = A(1-e\cos E)$ to get

$$\begin{aligned}
\frac{dE}{dt} &= \frac{r}{A\sqrt{1-e^2}}\frac{d\theta}{dt} \qquad \text{(B.6.7)} \\
&= \frac{A(1-e\cos E)}{A\sqrt{1-e^2}}\frac{d\theta}{dt} \qquad \text{(B.6.8)} \\
&= \frac{(1-e\cos E)}{\sqrt{1-e^2}}\frac{d\theta}{dt} \qquad \text{(B.6.9)}
\end{aligned}$$

Now we have an equation that links the rate of change of E to the rate of change of θ in terms of just E and θ

Using Angular Momentum given by equation B.3.2

$$l = mr^2\frac{d\theta}{dt}$$

useing equation B.5.27 and rearranging we obtain

$$\begin{aligned} l &= mr^2 \frac{d\theta}{dt} \\ \frac{d\theta}{dt} &= \frac{l}{mr^2} \\ \frac{d\theta}{dt} &= \frac{l}{mA^2 \left(1 - e\cos E\right)^2} \end{aligned} \tag{B.6.10}$$

which we can now substitute in for $\frac{d\theta}{dt}$ term in equation B.6.9

$$\begin{aligned} \frac{dE}{dt} &= \frac{(1 - e\cos E)}{\sqrt{1-e^2}} \frac{d\theta}{dt} \\ &= \frac{(1 - e\cos E)}{\sqrt{1-e^2}} \frac{l}{mA^2 \left(1 - e\cos E\right)^2} \\ &= \frac{l}{mA^2\sqrt{1-e^2}} \frac{1}{1 - e\cos E} \\ (1 - e\cos E) \frac{dE}{dt} &= \frac{l}{mA^2\sqrt{1-e^2}} \end{aligned} \tag{B.6.11}$$

Now you can see that the right hand side is just a constant, which is defined by properties of the orbit. Namely the angular momentum, the mass of the orbiting object, the semi major axis and the eccentricity. An interesting thing to note is that the term $A^2\sqrt{1-e^2}$ can be rewritten as the product AB instead, so linking the major and minor axes. We will however keep it in it's original form for now. Since the right hand side is a constant we will simplify by choosing the constant k for it. This constant k is the *Mean Motion*, later[11] we will see that it's connected to the period of the orbit, and ends up being the mean angular velocity of the orbiting body.

$$k = \frac{\frac{l}{m}}{A^2\sqrt{1-e^2}} \tag{B.6.12}$$

[11] see equation B.8.8 on page 59

so finally all that is left to do is to integrate our equation with respect to time.

$$\begin{aligned}
\int_{t_0}^{t} k dt &= \int_{t=t_0}^{t} (1 - e\cos E)\,\frac{dE}{dt} dt \\
k(t - t_0) &= \int_{E=0}^{E} (1 - e\cos E)\, dE \\
&= \int_0^E 1 dE - e\int_0^E \cos E dE \\
&= E - e\sin E && \text{(B.6.13)} \\
& && \text{(B.6.14)}
\end{aligned}$$

Note we have chosen t_0 to be the point in time when $E = 0$, which is the same as the statement of Kepler's equation.

$$E - e\sin E = k(t - t_0) \tag{B.6.15}$$

B.7 Eliptical Orbits

We can show that a general solution to the equations of motion is an elipse.

B.7.1 Change of variables

Firstly we start by letting $r = \frac{1}{u}$, so that

$$\begin{aligned}
\frac{dr}{dt} &= \frac{du^{-1}}{dt} \\
&= \frac{du^{-1}}{du}\frac{du}{dt} \\
&= -\frac{1}{u^2}\frac{du}{dt} \\
&= -r^2\frac{du}{dt}
\end{aligned} \tag{B.7.1}$$

next we introduce an ω by chain rule

$$\begin{aligned}
\frac{dr}{dt} &= -r^2\frac{du}{dt} \\
\frac{dr}{dt} &= -r^2\frac{du}{d\theta}\frac{d\theta}{dt} \\
&= -r^2\omega\frac{du}{d\theta}
\end{aligned}$$

And using the magnitude of angular momentum[12] $l = mr^2\omega$ or rearranging $\omega = \frac{l}{mr^2}$

[12] equation B.3.2

$$\begin{aligned}
\frac{dr}{dt} &= -r^2\omega\frac{du}{d\theta} \\
\frac{dr}{dt} &= -r^2\frac{l}{mr^2}\frac{du}{d\theta} \\
\frac{dr}{dt} &= -\frac{l}{m}\frac{du}{d\theta} \\
\frac{dr}{dt} &= -\frac{l}{m}\frac{du}{d\theta}
\end{aligned} \tag{B.7.2}$$

Differentiating again with respect to time gives

$$\begin{aligned}
\frac{dr^2}{dt^2} &= -\frac{l}{m}\frac{d}{dt}\frac{du}{d\theta} \\
&= -\frac{l}{m}\frac{d\theta}{dt}\frac{d}{d\theta}\frac{du}{d\theta} \\
&= -\frac{l}{m}\omega\frac{d^2u}{d\theta^2}
\end{aligned} \tag{B.7.3}$$

We can substitute the magnitude of angular momentum[13] $l = mr^2\omega$ again to get

$$\begin{aligned}
\frac{dr^2}{dt^2} &= -\frac{l}{m}\omega\frac{d^2u}{d\theta^2} \\
&= -\frac{mr^2\omega}{m}\omega\frac{d^2u}{d\theta^2} \\
&= -r^2\omega^2\frac{d^2u}{d\theta^2}
\end{aligned} \tag{B.7.4}$$

We can now substitute this into the radial part of the acceleration[14]

[13] equation B.3.2
[14] equation B.2.3

$$\begin{aligned}
-\frac{GM}{r^2} &= \frac{d^2r}{dt^2} - r\omega^2 \\
&= -r^2\omega^2\frac{d^2u}{d\theta^2} - r\omega^2 \\
&= -r^2\omega^2\left[\frac{d^2u}{d\theta^2} + \frac{1}{r}\right] \\
GM &= r^4\omega^2\left[\frac{d^2u}{d\theta^2} + \frac{1}{r}\right] \qquad \text{(B.7.5)}
\end{aligned}$$

Next we recall that $u = \frac{1}{r}$ and that $\frac{l}{m} = r^2\omega$ and substitute these in

$$\begin{aligned}
GM &= \left(\frac{l}{m}\right)^2\left[\frac{d^2u}{d\theta^2} + u\right] \\
\frac{d^2u}{d\theta^2} + u &= \frac{GMm^2}{l^2} \qquad \text{(B.7.6)}
\end{aligned}$$

To simplify this equation we will use $\alpha = \frac{GMm^2}{l^2}$

B.7.2 Solution

The direct solution to

$$\frac{d^2u}{d\theta^2} + u = \alpha \qquad \text{(B.7.7)}$$

is $u = \alpha$

However because there is also a function that results in zero, we must include that

$$\begin{aligned}
\frac{d^2u}{d\theta^2} + u &= 0 \\
\frac{d^2u}{d\theta^2} &= -u \qquad \text{(B.7.8)}
\end{aligned}$$

we can take the solution that $u = c\cos\theta$ which gives the general overall solution of

$$u(\theta) = \alpha + c\cos\theta \tag{B.7.9}$$

We can pick an initial condition when $\theta = 0$ the body is at periapsis or $r = p = A(1-e)$ which gives $u = \frac{1}{A(1-e)}$ (see the ellipse notes for $p = A(1-e)$)

$$\begin{aligned}\frac{1}{A(1-e)} &= \alpha + c \\ \alpha &= \frac{1}{A(1-e)} - c\end{aligned} \tag{B.7.10}$$

we could similarly hae picked the condition that when $\theta = \pi$ the body is at apoapsis or $r = a = A(1+e)$ which gives $u = \frac{1}{A(1+e)}$ (see ellipse notes for $a = A(1+e)$)

$$\begin{aligned}\frac{1}{A(1+e)} &= \alpha - c \\ \alpha &= \frac{1}{A(1+e)} + c\end{aligned} \tag{B.7.11}$$

we can equate our two initial conditions to get c in terms of just a and p

$$
\begin{aligned}
\alpha = \frac{1}{A(1-e)} - c &= \frac{1}{A(1+e)} + c \\
\frac{1}{A(1-e)} - \frac{1}{A(1+e)} &= 2c \\
\frac{A(1+e) - A(1-e)}{A^2(1+e)(1-e)} &= 2c \\
\frac{A + eA - A + eA}{A^2(1-e^2)} &= 2c \\
\frac{2eA}{A^2(1-e^2)} &= 2c \\
\frac{e}{A(1-e^2)} &= c
\end{aligned}
\tag{B.7.12}
$$

Now that we have c we can use this to determine α in terms of A and e

$$
\begin{aligned}
\alpha &= \frac{1}{A(1+e)} + c \\
&= \frac{1}{A(1+e)} + \frac{e}{A(1-e)^2} \\
&= \frac{(1-e)}{A(1+e)(1-e)} + \frac{e}{A(1-e)^2} \\
&= \frac{1-e+e}{A(1-e^2)} \\
&= \frac{1}{A(1-e^2)}
\end{aligned}
$$

So feeding all this back in we have

$$\begin{aligned} u(\theta) &= \alpha + c\cos\theta \\ &= \frac{1}{A(1-e^2)} + \frac{e}{A(1-e^2)}\cos\theta \\ &= \frac{1+e\cos\theta}{A(1-e^2)} \end{aligned} \tag{B.7.13}$$

Now we have a simple form for u we can recall that we chose $u = \frac{1}{r}$, so

$$r = \frac{1+e\cos\theta}{A(1-e^2)} \tag{B.7.14}$$

which is the equation for a ellipse[15].

We also get the interesting result here that

$$\begin{aligned} \alpha = \frac{GMm^2}{l^2} &= \frac{1}{A(1-e^2)} \\ A(1-e^2) &= \frac{l^2}{GMm^2} \end{aligned} \tag{B.7.15}$$

which connects the semi major axis and the eccentricity to physical properties, like the masses and the angular momentum. This will help us to work out the geometry from the physics.

[15] equation B.5.23

B.8 Relationship between geometry and physics

We saw in the previous chapter that the geometry and the physics were linked by equation B.7.15.

B.8.1 Angular Momentum From Geometry

We can rearrange equation B.7.15 to get

$$\begin{aligned} \frac{l^2}{GMm^2} &= A(1-e^2) \\ \left(\frac{l}{m}\right)^2 &= GMA(1-e^2) \end{aligned} \tag{B.8.1}$$

Which can be further rearranged using equations B.5.12[16] and B.5.13[17]

$$\begin{aligned} \left(\frac{l}{m}\right)^2 &= GM\frac{A^2(1-e^2)}{A} \\ &= GM\frac{B^2}{A} \\ &= GM\frac{ap}{A} \\ &= GM\frac{2ap}{a+p} \end{aligned} \tag{B.8.2}$$

With some further rearrangement we can obtain the angular momentum per unit mass of the orbiting body

[16] page 41
[17] page 41

$$\begin{aligned}
\frac{l^2}{GMm^2} &= \frac{2ap}{a+p} \\
\left(\frac{l}{m}\right)^2 &= GM\frac{2ap}{a+p} \\
\frac{l}{m} &= \sqrt{GM\frac{2ap}{a+p}}
\end{aligned} \tag{B.8.3}$$

B.8.2 Total Energy From Geometry

The total energy[18] is given by

$$E(r,\dot{r}) = \frac{1}{2}m\dot{r}^2 + \frac{l^2}{2mr^2} - \frac{GMm}{r}$$

making this total energy per unit mass of the orbiting body instead

$$\frac{E}{m} = \frac{1}{2}\dot{r}^2 + \frac{1}{2r^2}\left(\frac{l}{m}\right)^2 - \frac{GM}{r} \tag{B.8.4}$$

[18] eqation B.4.6

If we choose one of the apses, say the periapsis we know that at this point $\dot{r} = 0$ so the total energy per unit mass is

$$\begin{aligned}
\frac{E}{m} &= \frac{1}{2p^2}\left(\frac{l}{m}\right)^2 - \frac{GM}{p} \\
&= \frac{1}{2p^2}\left(GM\frac{2ap}{a+p}\right) - \frac{GM}{p} \\
&= GM\left[\frac{2ap}{2p^2(a+p)} - \frac{1}{p}\right] \\
&= GM\left[\frac{2ap}{2p^2(a+p)} - \frac{2p(a+p)}{2p^2(a+p)}\right] \\
&= GM\left[\frac{2ap - 2p(a+p)}{2p^2(a+p)}\right] \\
&= GM\left[\frac{-2p^2}{2p^2(a+p)}\right] \\
&= -GM\left[\frac{1}{a+p}\right] \\
&= -\frac{GM}{2}\left[\frac{2}{a+p}\right] \\
&= -\frac{GM}{2A}
\end{aligned} \tag{B.8.5}$$

Note that energy is negative because we took zero to be when the particle is at infinity. We can rearrange equation B.8.5 to obtain the semi major axis as

$$\begin{aligned}
\frac{E}{m} = \tilde{E} &= -\frac{GM}{2A} \\
2A\tilde{E} &= -GM \\
A &= -\frac{GM}{2\tilde{E}}
\end{aligned} \tag{B.8.6}$$

Where we are using $\tilde{E} = \frac{E}{m}$ to be the energy per unit mass of the orbiting body. This can then be fed back into our relationship for angular momentum[19].

$$
\begin{aligned}
\tilde{l}^2 &= GMA(1-e^2) \\
&= -\frac{(GM)^2(1-e^2)}{2\tilde{E}} \\
-\frac{2\tilde{E}\tilde{l}^2}{(GM)^2} &= 1-e^2 \\
e^2 - \frac{2\tilde{E}\tilde{l}^2}{(GM)^2} &= 1 \\
e^2 &= 1 + \frac{2\tilde{E}\tilde{l}^2}{(GM)^2} \\
e &= \sqrt{1 + \frac{2\tilde{E}\tilde{l}^2}{(GM)^2}} \qquad \text{(B.8.7)}
\end{aligned}
$$

Which means we can work out the semi major axis and the ecentricity based on the physics.

B.8.3 Period

An orbiting body will complete a complete orbit when it has gone through an angle $E = 2\pi$, we can then use Kepler's Equation[20] to compute the period. For ease of calculation we will assume that $t_0 = 0$.

$$
\begin{aligned}
E - e\sin E &= kt \\
2\pi - e\sin(2\pi) &= kT \\
T &= \frac{2\pi}{k} \qquad \text{(B.8.8)}
\end{aligned}
$$

[19]equation B.8.2
[20]equation B.6.15

The Relationship between Mean Motion and Time Period

As we have seen from equation B.8.8, the mean motion is related to the orbital period

$$k = \frac{2\pi}{T} \tag{B.8.9}$$

which is how the constant gets the name *Mean Motion* as it is the mean angular velocity of the orbiting body. In this form Kelper's Equation[21] becomes

$$E - e\sin E = 2\pi\frac{(t - t_0)}{T} \tag{B.8.10}$$

Period based on physical and geometric properties

The period can be related to other geometric and physical properties by substituting equation B.6.12 in for k, we get

$$T = \frac{2\pi A^2\sqrt{1 - e^2}}{\frac{l}{m}} \tag{B.8.11}$$

We can use equation B.8.2[22] to replace $\frac{l}{m}$.

$$\begin{aligned} T &= \frac{2\pi A^2\sqrt{1 - e^2}}{\frac{l}{m}} \\ &= \frac{2\pi A^2\sqrt{1 - e^2}}{\sqrt{GMA(1 - e^2)}} \\ &= \frac{2\pi A^2\sqrt{1 - e^2}}{\sqrt{GM}\sqrt{A}\sqrt{1 - e^2}} \\ &= \frac{2\pi\sqrt{A^3}}{\sqrt{GM}} \end{aligned} \tag{B.8.12}$$

[21] equation B.6.15 on page 49

[22] page 56

We can get this in terms of the apses by using equation B.5.4 on page 40 to substitute in for $A = \frac{a+p}{2}$.

$$
\begin{aligned}
T &= \frac{2\pi\sqrt{A^3}}{\sqrt{GM}} \\
T^2 &= \frac{4\pi^2}{GM}A^3 \\
&= \frac{4\pi^2}{GM}\left(\frac{a+p}{2}\right)^3
\end{aligned}
\tag{B.8.13}
$$

Note that we have squared T to make the equation easier to work with, but this reproduces one of Kepler's Law, that the square of the period is proportional to the cube of the semi major axis[23].

[23] or more usefully the average of the apses

B.9 In Plane Position and Velocity over time

We will assume that the apses are know as well as the central mass. From this the period is given by equation B.8.13 on page 61

$$T = \sqrt{\frac{4\pi^2}{GM}} \left(\frac{a+p}{2}\right)^{\frac{3}{2}}$$

At any given time we can calculate the ecentric anomaly E using Kepler's Equation[24]

$$E - e\sin E = 2\pi\frac{(t-t_0)}{T}$$

Since this is trancendental, it will need to be solved numerically, or by approximation. This text will not cover those methods, and assume that E can be determined as a function of t.

B.9.1 Position

Once E is known, then the position can de determined by

$$\begin{aligned} e &= \frac{a-p}{a+p} \\ r &= A\left(1 - e\cos E\right) \\ x &= r\cos\theta = A(\cos(E) - e) \\ y &= r\sin\theta = A\sqrt{1-e^2}\sin E \end{aligned}$$

B.9.2 Velocitiy

Angular Velocity

Using equation B.8.2 on page 56 to get the angular momentum

$$\left(\frac{l}{m}\right) = \sqrt{GM}\sqrt{\frac{2ap}{a+p}}$$

[24] equation B.8.10 on page 60

and using equation ?? on page ??, we can get the angular velocity as

$$\omega = \frac{l}{m}\frac{1}{r^2} = \sqrt{GM}\sqrt{\frac{2ap}{a+p}}\frac{1}{r^2} \tag{B.9.1}$$

Radial velocity

The radial velocity we obtain from the geometry. Recall that we have

$$r = A(1 - e\cos E) \tag{B.9.2}$$

As we did for the derivation of Kepler's Equation, we can differentiate with respect to time. Which gives the result

$$\frac{dr}{dt} = eA\sin E\frac{dE}{dt} \tag{B.9.3}$$

And looking at Kepler's Equation again

$$\begin{aligned} E - e\sin E &= 2\pi\frac{t-t_0}{T} \\ t &= \frac{E - e\sin E}{2\pi}T + t_0 \\ \frac{dt}{dE} &= \frac{1 - e\cos E}{2\pi}T \end{aligned}$$

which gives us the result that

$$\frac{dE}{dt} = \frac{2\pi}{T}\frac{1}{1 - e\cos E} \tag{B.9.4}$$

We can put this back in to our earlier equation to get the radial velocity

$$\frac{dr}{dt} = \frac{2\pi}{T}\frac{eA\sin E}{1 - e\cos E} \tag{B.9.5}$$

or in terms of the apses

$$\frac{dr}{dt} = \frac{2\pi}{T}\frac{a-p}{2}\frac{\sin E}{1 - e\cos E} \tag{B.9.6}$$